

KINGSVILLE NAS, TX, USA

WMO: 722516

Lat: 27.500N

Lon: 97.817W

Elev: 15

StdP: 101.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	2.9	-8.3	1.9	8.8	-5.4	2.4	10.3	12.5	23.3	11.3	20.0	3.7	0	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	37.3	25.8	36.3	25.8	35.5	25.8	28.0	33.3	27.5	32.4	27.1	31.8	5.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.6	29.3	26.4	21.9	28.9	26.1	21.5	28.8	89.5	33.7	87.2	32.4	85.3	31.7	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.9	8.9	DB	-2.2	39.5	2.2	1.9	-3.8	40.9	-5.1	42.0	-6.3	43.0	-7.9	44.4	(4)
			WB	-3.7	28.9	2.0	0.8	-5.1	29.5	-6.3	30.0	-7.4	30.4	-8.9	31.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.0	14.2	16.7	19.5	23.2	26.6	29.1	29.8	30.0	27.7	23.9	18.9	15.5
	DBStd	6.79	5.32	5.46	4.85	3.70	2.66	1.73	1.40	1.51	2.27	3.76	4.96	5.36
	HDD10.0	49	21	9	3	0	0	0	0	0	0	0	3	12
	HDD18.3	469	148	87	47	7	0	0	0	0	0	6	53	123
	CDD10.0	4777	152	197	297	397	516	572	614	620	531	429	270	182
	CDD18.3	2156	21	41	82	154	258	322	356	362	281	177	69	34
	CDH23.3	22882	145	277	507	1238	2596	3850	4432	4452	2938	1682	553	212
CDH26.7	10455	26	77	133	430	1052	1911	2307	2337	1344	647	150	41	
Wind	WSAvg	4.3	4.2	4.5	4.8	5.0	5.1	4.6	4.7	4.1	3.3	3.7	4.0	4.0
Precipitation	PrecAvg	712	36	41	23	40	73	85	59	77	109	81	32	35
	PrecMax	1262	96	105	117	209	288	316	305	532	492	347	128	220
	PrecMin	323	0	0	0	0	0	2	0	1	8	2	0	0
	PrecStd	259	24	32	26	46	64	68	66	95	98	83	31	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	32.1	32.5	35.8	35.9	37.8	38.0	38.6	37.2	34.3	32.3	29.7
		MCWB	19.8	20.1	21.8	23.4	25.1	25.8	25.9	25.6	24.8	24.9	23.6	21.5
	2%	DB	27.0	28.8	29.7	32.7	34.1	36.1	36.8	37.4	35.6	32.9	30.1	27.8
		MCWB	19.4	20.8	21.7	22.6	25.1	26.1	25.9	26.1	25.5	24.8	22.7	20.9
	5%	DB	24.8	26.8	27.9	30.8	32.9	35.1	36.0	36.5	34.1	31.7	28.3	26.0
		MCWB	18.6	20.0	21.3	22.8	24.9	25.9	26.0	26.1	25.1	24.3	22.1	20.3
	10%	DB	22.7	24.5	26.3	29.0	31.8	34.0	34.9	35.4	32.8	30.4	26.6	23.7
		MCWB	18.0	19.1	20.3	22.4	24.7	25.6	26.0	25.8	25.2	23.6	21.3	20.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	23.1	24.5	26.8	28.0	28.7	28.6	28.7	27.9	27.3	25.6	23.7
		MCDB	25.5	27.7	29.0	30.4	32.0	33.9	34.9	34.9	32.2	30.9	29.0	27.0
	2%	WB	21.5	22.3	23.5	25.4	26.9	27.9	27.9	27.8	27.2	26.5	24.6	22.7
		MCDB	24.2	26.3	27.5	29.6	30.8	33.0	33.2	33.3	31.5	29.9	27.6	25.5
	5%	WB	20.6	21.5	22.7	24.5	26.3	27.3	27.3	27.2	26.8	25.8	23.7	21.8
		MCDB	23.4	24.8	26.0	28.4	30.3	32.2	32.2	32.4	30.9	29.2	26.6	24.3
	10%	WB	19.4	20.7	21.9	23.8	25.7	26.8	26.9	26.8	26.3	25.2	22.5	20.9
		MCDB	22.1	23.2	24.4	27.3	29.6	31.3	31.7	31.8	30.3	28.7	25.5	23.3
Mean Daily Temperature Range	5% DB	MDBR	12.1	11.8	11.1	11.2	10.1	10.4	10.9	11.2	10.6	11.8	11.9	11.4
		MCDBR	14.2	14.0	12.0	12.3	11.1	11.5	12.2	12.6	12.1	11.8	12.6	13.1
		MCWBR	8.3	7.1	5.8	5.4	3.8	3.3	3.3	3.2	3.7	5.0	6.3	7.4
	5% WB	MCDBR	11.5	11.3	10.0	9.9	8.9	9.7	10.6	11.2	10.1	9.8	10.9	10.8
		MCWBR	7.5	6.2	5.3	4.6	3.8	3.6	3.4	3.4	3.6	4.8	6.3	7.1
Clear-Sky Solar Irradiance	taub	0.358	0.368	0.385	0.431	0.470	0.447	0.453	0.449	0.439	0.387	0.374	0.351	
	taud	2.455	2.429	2.384	2.267	2.190	2.314	2.316	2.324	2.320	2.456	2.442	2.489	
	Ebn at Noon	885	905	907	870	832	846	840	843	843	873	860	877	
	Edh at Noon	97	108	119	138	149	131	130	128	125	104	98	91	
All-Sky Solar Radiation	RadAvg	3.06	3.65	4.58	5.45	6.00	6.66	6.61	6.34	5.22	4.54	3.42	2.75	
	RadStd	0.33	0.61	0.54	0.53	0.48	0.31	0.52	0.35	0.45	0.49	0.36	0.39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.44	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	+0.35	N/A	N/A	N/A	N/A	+121	+112

Nomenclature: See separate page